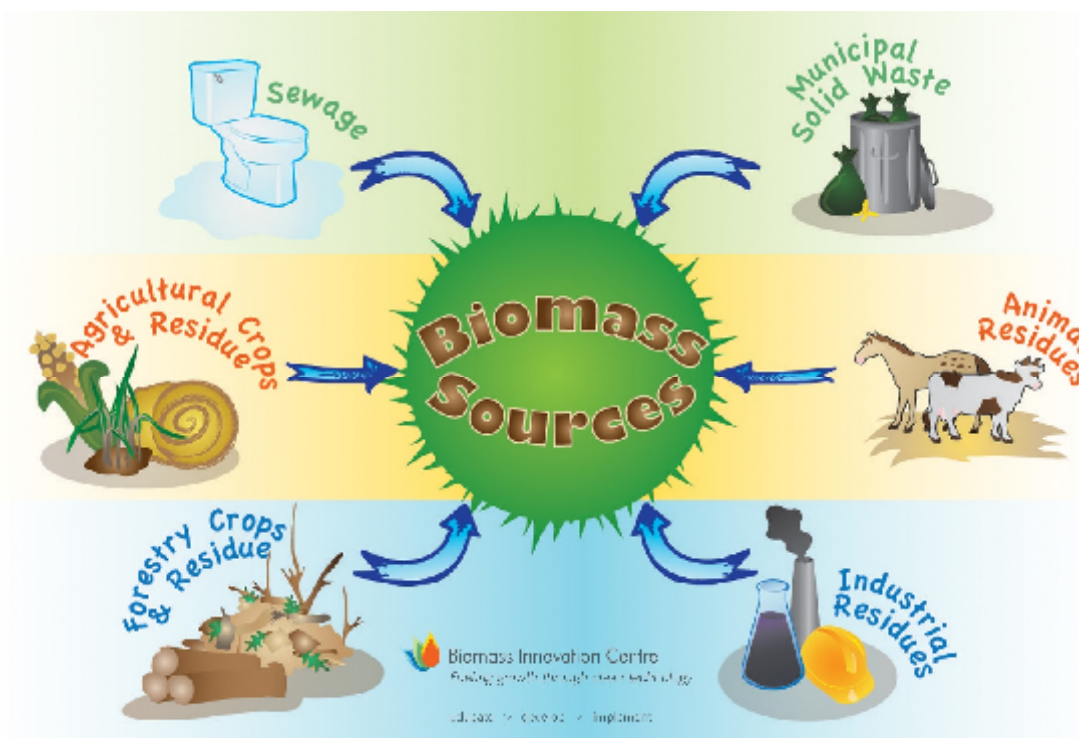


MODULE -4

BIOMASS ENERGY

Session 25

Biomass is renewable organic material that comes from plants and animals. Biomass contains stored chemical energy from the sun that is produced by plants through photosynthesis. Biomass can be burned directly for heat or converted to liquid and gaseous fuels through various processes.



9.1 BIOMASS PRODUCTION

Organic substances exist in wide variety from living beings to dead organisms. Organic matters are composed of Carbon (C), combined with elements such as Hydrogen (H), Oxygen (O), Nitrogen (N), Sulphur (S) to form variety of organic compounds such as carbohydrates, proteins & lipids. In nature MOs (microorganisms), through digestion process breaks the complex carbon into smaller substances.



9.1.1 Direct method

Raw materials that can be used to produce biomass energy are available throughout the world in the following forms:

1. Forest wood and wastes
2. Agricultural crops and residues

3. Residential food wastes
4. Industrial wastes
5. Human and animal wastes
6. Energy crops

Raw biomass has a low energy density based on their physical forms and moisture contents and their direct use are burning them to produce heat for cooking. The twin problems of traditional biomass use for cooking and heating are the energy inefficiency and excessive pollution

9.1.2 Indirect method

1. **Thermo electrical conversion:** The direct combustion of biomass material in the boiler produces steam that is used either to drive a turbine coupled with an electrical generator to produce electricity or to provide heat for residential and industrial system
2. **Biomass conversion to fuel:** Biomass conversion processes can be classified under two main types: (a) Thermo-chemical conversion includes processes such as destructive distillation, pyrolysis, and gasification. (b) Biological conversion includes processes such as fermentation and anaerobic digestion

Gasification produces a synthesis gas with usable energy content by heating the biomass with less oxygen than needed for complete combustion. Pyrolysis yields bio-oil by rapidly heating the biomass in the absence of oxygen. Anaerobic digestion produces a renewable natural gas (methane gas) when organic matter is decomposed by bacteria in the absence of oxygen. As a result, it is often advantageous to convert this waste into more readily usable fuel form like producer gas. Hence, it is the attractiveness of gasification. The efficiency of a direct combustion or biomass gasification system is influenced by several factors such as including biomass moisture content, combustion air distribution and amounts (excess air), operating temperature and pressure, and flue gas (exhaust) temperature.

9.2 ENERGY PLANTATION

An interesting approach for the large-scale planned use of wood is the ‘energy plantation’ approach. energy plantation means growing select species of trees and shrubs which are harvestable in a comparably shorter time and are specifically meant for fuel. The fuel wood may be used either directly in wood burning stoves and boilers or processed into methanol, ethanol, and producer gas.

It has been suggested that electrical power be produced by the energy plantation approach, the wood grown in this manner being used as a fuel for the boilers of a conventional power plant. The technology of biomass-based electric power plants is well established in the USA and Europe and there are over 500 such plants use wood, wood waste, and various types of agricultural waste.

9.3 BIOMASS GASIFICATION

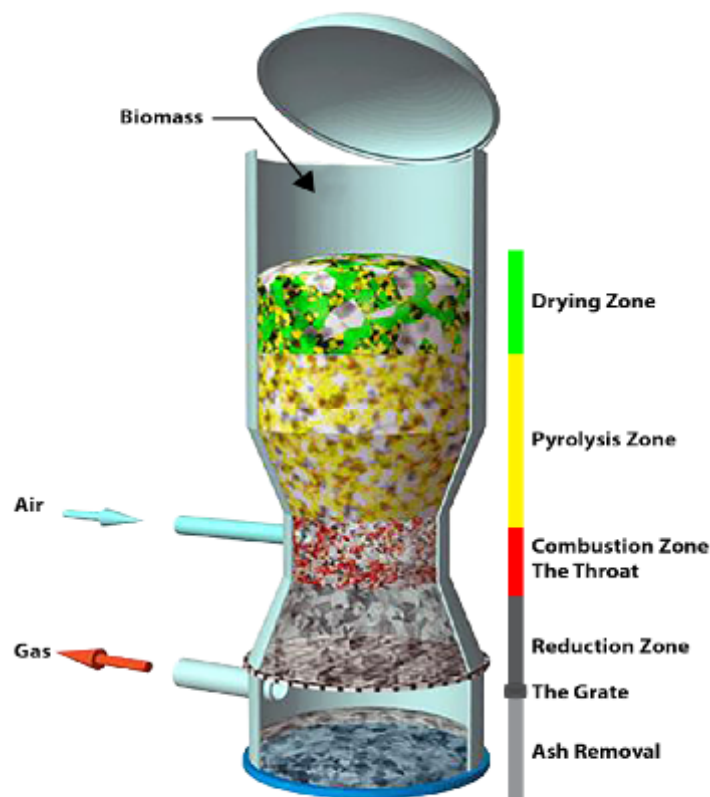
Gasification is **conversion of solid Hydrocarbaceous fuels** (wood/ wood-waste, agricultural residues, chicken manure, coal, Municipal Waste etc.) into a combustible gas mixture called **Producer Gas/ Syn Gas**. The Gasifier is essentially a chemical reactor where various physical and chemical processes take place and break the solid fuel down into Producer Gases. Four distinct processes take place in a gasifier:

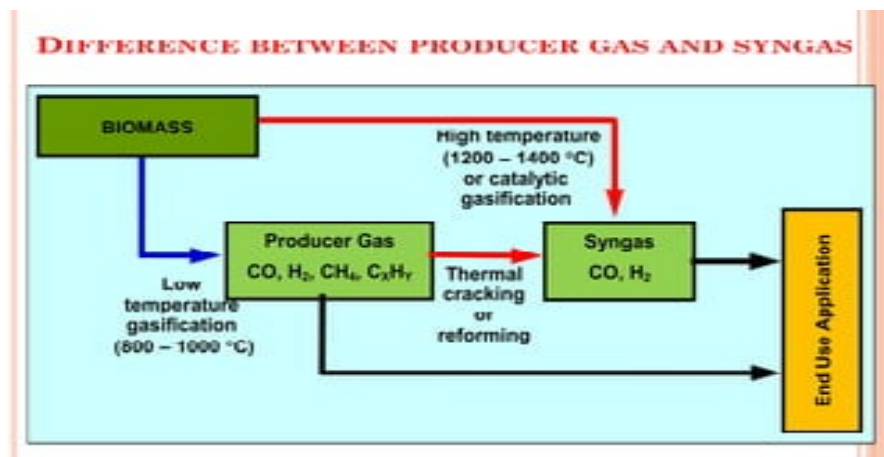
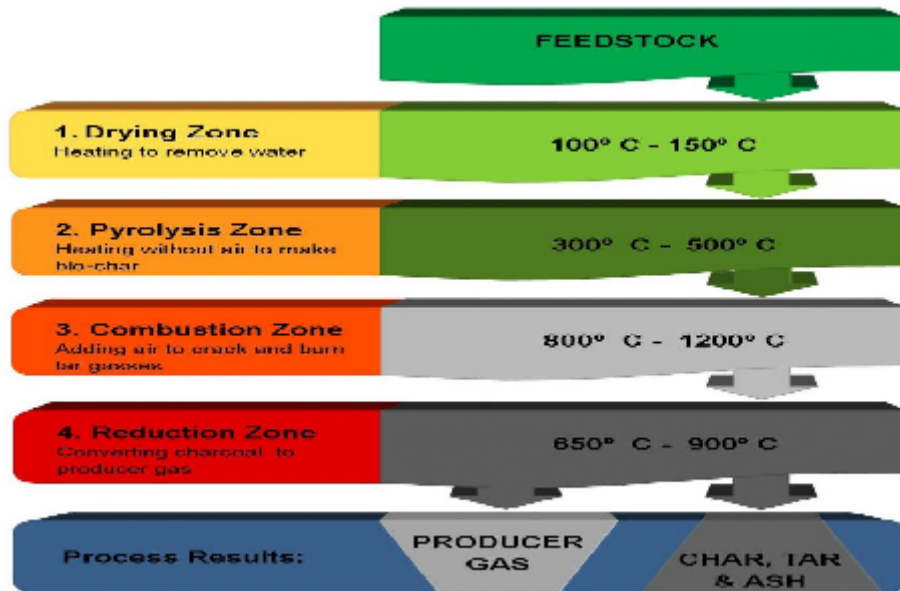
1. Drying of the fuel: The moisture in the feed comes out in this zone in the form of water vapor. Drying takes place in the upper most portion of the Gasifier, through heat transferred from the high temperature combustion zone.

2. Pyrolysis This is chemical decomposition of organic materials by heating in absence of oxygen at temperatures above 200 degrees C. During pyrolysis, volatiles are released (in the form of gas) and char is produced.

3. Combustion This is where controlled oxygen is given to the fuel and oxidation/ burning occurs. Heat and energy are released.

4. Reduction The combustion products mainly CO_2 (Carbon-di-Oxide) and H_2O (water vapor) get reduced in the presence of high temperature carbon to finally give CO (Carbon-mono-oxide) and H_2 (Hydrogen).





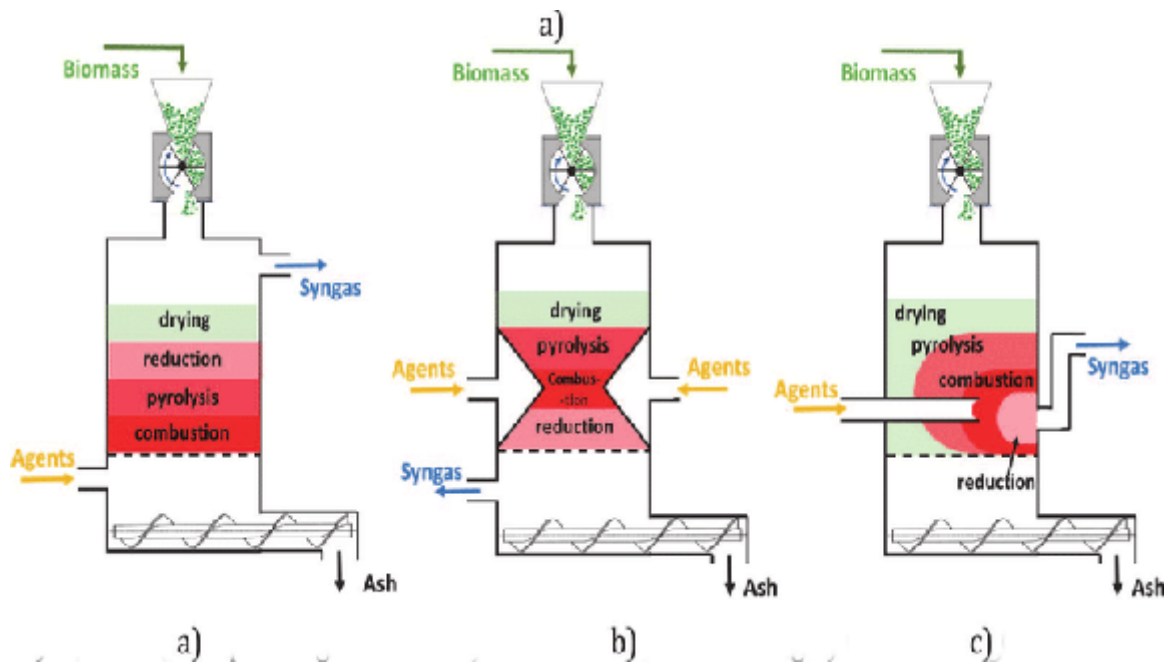
GASIFIER AND THEIR CLASSIFICATIONS

Biomass gasifier may be considered as a chemical reactor in which biomass goes through several complex physical and chemical processes and producer or syngas is produced and recovered.

There are two distinct types of gasifiers:

1. **Fixed bed gasifier**: In this gasifier, biomass fuels move either counter current or concurrent to the flow of gasification medium (steam, air, or oxygen) as the fuel is converted to fuel gas. They are relatively simple to operate and have reduced erosion.

Since there is an interaction of air or oxygen and biomass in the gasifier, they are classified according to the way air or oxygen is introduced in it.



- (a) **Downdraft gasifiers:** In the downdraft gasifier, the air is passed from the layers in the downdraft direction. Single throat gasifiers are mainly used for stationary applications, whereas double throat gasifier is used for varying loads as well as automotive purposes.
- (b) **Updraft gasifiers:** Updraft gasifier has air passing through the biomass from bottom and the combustible gases come out from the top of the gasifier.
- (c) **Cross draft gasifiers:** It is a very simple gasifier and is highly suitable for small outputs. With slight variation, almost all the gasifiers fall in the abovementioned categories.

2. **Fluidized bed gasifier:** In fluidized bed gasifier, an inert material (such as sand, ash, or char) is utilized to make bed and that acts as a heat transfer medium

CHEMISTRY OF REACTION PROCESS IN GASIFICATION

_ Drying Zone of fuel- the moisture content of biomass is removed off,

200°C-280°C carbon dioxide, acetic acid and water,

280°C-500°C Large quantity of tar,

500°C-700°C gas is produced.

_ Combustion zone :

$C + O_2 \rightarrow CO_2 + \text{Heat}$

_ Reduction Zone

endothermic reaction- $C + CO_2 + \text{heat} \rightarrow 2CO$

Gas Shift reaction $C + H_2O + \text{heat} \rightarrow CO + H_2$

Exothermic reaction- $CO + H_2O \rightarrow \text{Heat} + CO_2 + H_2$

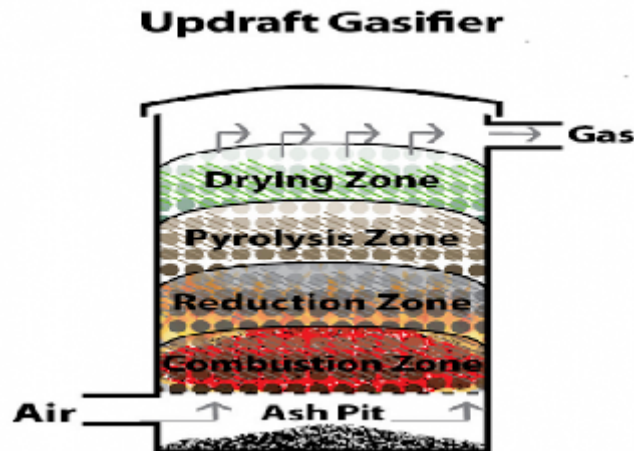
Review questions:

1. Define biomass

2. Define biomass gasification
3. Types of gasifiers.

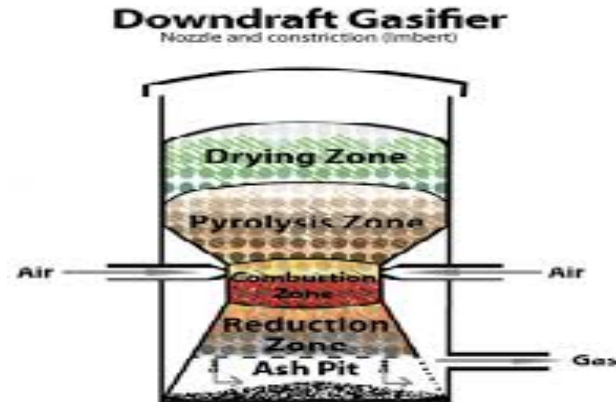
Session 26

9.7 UPDRAFT GASIFIERS



The updraft fixed bed ("counter-current") gasifier consists of a fixed bed of carbonaceous fuel (e.g. coal or biomass) through which the "gasification agent" (steam, oxygen, and/or air) flows in counter-current configuration. The ash is either removed dry or as a slag. The updraft gasifier consists of a top fed fuel bed through which the "gasification agent" (steam, oxygen, and/or air) flows in from the bottom and exits through the top as gas. Updraft gasifiers are thermally efficient because the ascending gases pyrolyze and dry the incoming biomass, transferring heat so that the exiting gases leave very cool.

The updraft gasifier has been the standard of coal gasification for 150 years and it is also popular in biomass cook stoves.

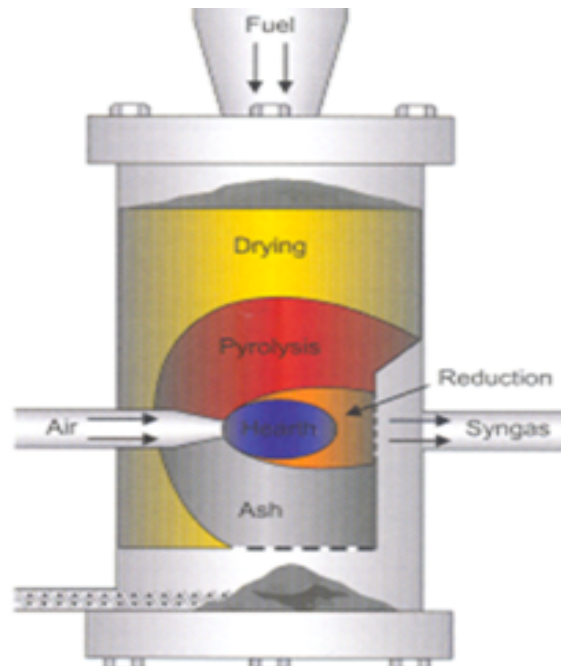


A downdraft gasifier is a co-current reactor where air enters the gasifier at a certain height below the top. The product gas flows downward (giving the name *downdraft*) and leaves through a bed of hot ash. Since it passes through the high-temperature zone of hot ash, the tar in the product gas finds favourable conditions for cracking. For this reason, a downdraft gasifier, of all types, has the lowest tar production rate.

Downdraft gasifiers are widely used in the following applications:

1. Continuous baking ovens (bread, biscuits, and paint)
2. Batch type baking oven (rotary oven for bread)
3. Dryers and curing (tea, coffee, mosquito coil, and paper drying)
4. Boilers
5. Thermal fluid heaters
6. Annealing furnaces
7. Direct fired rotary kilns
8. Internal combustion engines

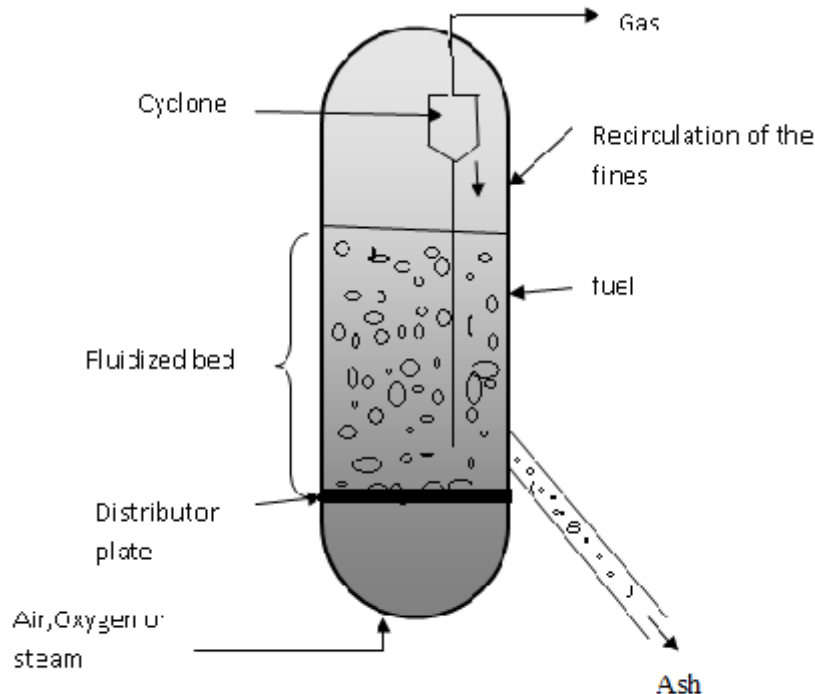
Cross draft gasifiers:



The gasifier is a vertical cylindrical vessel of varying cross section. The biomass is fed in at the top at regular intervals of time and is converted through a series of processes into producer gas and ash, as it moves down slowly through various zones of the gasifier. The crucial difference is that the air will be entering the gasifier from the side of the reactor, rather than from the top or the bottom

FLUIDIZED BED GASIFICATION

Fluidized bed gasification has been successfully used to convert prepared wastes (i.e., wood wastes, bark, agricultural wastes, and RDF (Refused Derived Fuel) into a clean fuel gas that can be used to fire various types of industrial equipment. The bed made of an inert material (such as sand, ash, or char) initially and it is heated and the fuel is introduced when the temperature has reached the appropriate level. plate at a controlled rate. Fluidized bed gasifiers have no distinct reaction zones (as in the case of fixed bed gasifiers) and drying, pyrolysis and gasification occur simultaneously. The fuel particles are introduced at the bottom of the reactor, very quickly mixed with the bed material, and almost instantaneously heated up to the bed temperature. As a result of this treatment, the fuel is pyrolyzed very fast, resulting in a component mix with a relatively large number of gaseous materials. Further gasification and tar-conversion reactions occur in the gas phase. For biomass feeds which have high ash content and the ash has low melting point, fluidized bed combustion seems to gasify them.



Advantages and Benefits

Advantages

1. Reduced cost of boiler or dryer operation by using wood and/or bark wastes rather than gas or oil.
2. Reduced cost for additional steaming capacity when compared to new wood and or barkfired boilers.
3. Reduced dependency on external fuel sources for propane, natural gas, and oil.

Benefits

1. **High overall efficiency:** High efficiency in the range of 70%–90% can be achieved.
2. **Fuel flexibility:** The fluidized bed gasifiers have fuel flexibility and operate satisfactorily with highly variable feed materials. Ranging from coal, shredded wood and bark to sawdust fines, or lump wood with particle sizes of less than 4–6 cm.
3. **Highly reliable:** The fluidized bed gasifier neither have moving grates nor other moving parts in the high temperature regions of the bed and hence they are highly reliable.
4. **Low purchase and installation costs:** Air flow used in the gasifiers is comparatively low, and hence size of gasifier is small and compact. These permit systems to be completely shop fabricated and assembled on skids, thereby reducing purchase price and installed costs.
5. **Flexible operations:** Fuel gas product of fluidized bed gasifier is easily applied to a variety of industrial processes including boilers, dry kilns, veneer dryers, or several pieces of equipment at once. Thus, they provide flexible operations.
6. **Low emissions:** They are very low emission gasifiers and do not require exhaust clean up devices.

USE OF BIOMASS GASIFIER

The output of a biomass gasifier can be used for a variety of **direct thermal applications** such as **cooking, drying, heating water, and generating steam**. It can also be used as a fuel for internal combustion engines to obtain mechanical **shaft power or electrical power**. If used as a fuel for internal combustion engines, it must be cleaned first for complete removal of particulate material and tar. **A cleaning system consisting of cyclone, a scrubber, and a filter is used for the purpose**. If the engine is a spark-ignition engine, it can operate with producer gas alone

Liquid Fuels

When compared to gaseous fuels such as producer gas or biogas, liquid fuels are somewhat harder to obtain from biomass sources. One of the methods is the production of **methanol** from **wood or straw**. The process involves the gasification of plant matter followed by chemical synthesis. Another method is the conversion of certain **food grains and crops such as sugarcane, maize and tapioca by fermentation into ethanol**. When blended with petrol,

ethanol is good alternate fuel for automotive engines. This fact has received considerable attention as a means of overcoming the oil crisis. However, if one examines the requirement of land for growing the agricultural products concerned, it is obvious that the method can be of substantial benefit only to a country having a large surplus of land. For this reason, Brazil has adopted this method on a large scale and produces significant amounts of ethanol for use as an alternate fuel. However, the position in India is quite different since the availability of land is limited.

GASIFIER BIOMASS FEED CHARACTERISTICS

Following biomass feed characteristics or parameters dictate the quality and classification of gasifiers:

1. Energy content of the fuel

The higher the energy content and bulk density of fuel, the smaller is the gasifier volume; as for one biomass fuel charge, power can be obtained for longer time duration.

2. Bulk density

3. Moisture content

It is desirable to use fuel with low moisture content to minimize heat loss due to its evaporation. Generally, desirable moisture content for fuel should be less than 20%.

4. Dust content

The gasifier design should be such that it should not produce dust beyond specified limits.

5. Tar content

Tar is one of the most unpleasant constituents of the gas as it tends to deposit in the carburettor and intake valves causing sticking and troublesome operations. It is a product of highly irreversible process taking place in the pyrolysis zone. Thus, the major effort has been devoted to cleaning this tar by filters and coolers.

6. Ash and slogging characteristic

The mineral content in the fuel that remains in oxidized form after complete combustion is usually called ash. Ash and tar removal are the two most important processes in gasification system for its smooth running.

Biomass Feed (Fuel)

The major biomass sources presently used are as follows:

1. Sugarcane and corn, wheat, sugar beet, sweet sorghum, and cassava to produce bioethanol.
2. Rapeseed, sunflower seeds, soybean, canola, peanuts, jatropha, coconut, and palm oil for biodiesel production.
3. Wide range of cellulosic materials (such as grassy crops, woody plants, by-products from the forestry and agricultural sector including wood residues, stems, and stalks and municipal wastes constitute the so-called second generation of feedstock).
4. Wastes and residues constitute a large source of biomass. These include solid and liquid municipal wastes, manure, lumber, and pulp mill wastes, and forest and agricultural residues.

Review questions:

1. What is updraft gasifier
2. What is downdraft gasifier
3. What is fluidized bed gasification.

Session 27

APPLICATION OF BIOMASS GASIFIER

1. Motive power
2. Direct heat application
3. Electrical power generation
4. Chemical production

COOLING AND CLEANING OF GASIFIER

For efficient and effective use of gas for numerous applications, it should be cleaned of tar and dust, free from moisture content and cooled. Therefore, cooling and cleaning of the gas is one of the most important processes in the whole gasification system. The failure or the success of producer gas units depends completely on their ability to provide a clean and cool gas to the engines or for burners. The temperature of gas coming out of generator is normally between 300°C and 500°C. The energy density of gas can be increased primarily by cooling it.

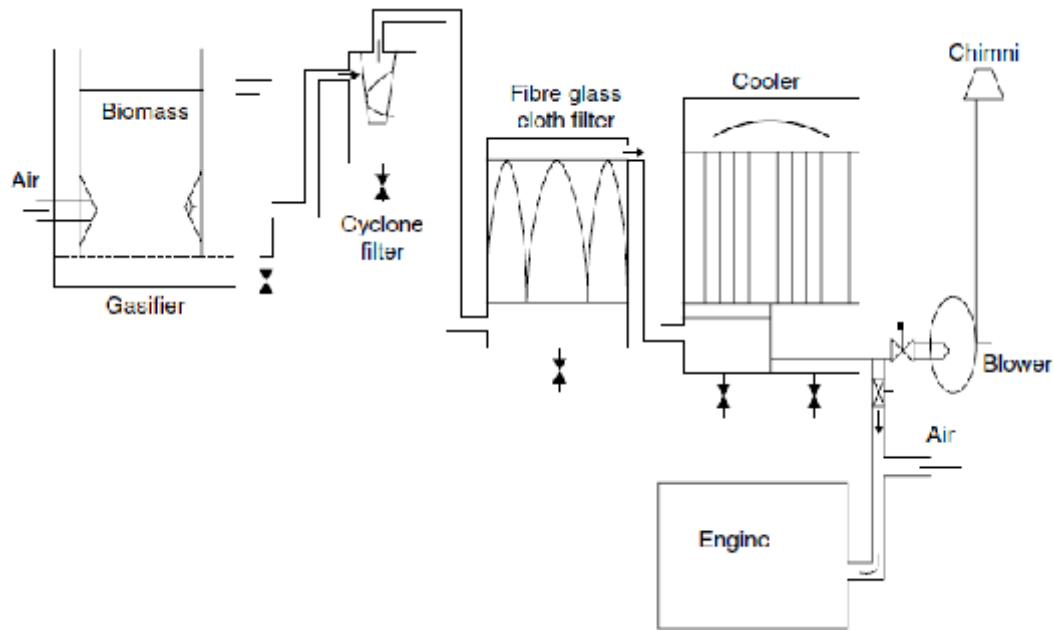
Normally, there are three types of filters used for cleaning of gas which is schematically a downdraft gasification system with cleaning and cooling train.

They are classified as dry, moist, and wet.

1. **Cyclone filters:** Cyclone dust collectors **can be used as pre-separators to reduce the dust load reaching a final, more efficient filter.** They can act as protective devices to remove large hot particles from the gas or air stream to prevent damage to the filter media/material.

2. **Wet scrubber:** Even after cyclone filtering, the gas still contains fine dust, particles, and tar. It is further cleaned by passing through a wet scrubber where gas is washed by water in Counter current mode. The scrubber also acts like a cooler, from where the gas goes to cloth filter for final cleaning.

3. **Cloth filters:** It is a fine filter in quite a few gasification systems, the hot gases are passed through the cloth filter, and then only do they go to the cooler.



Schematic diagram of producer gas plant

Biogas Energy

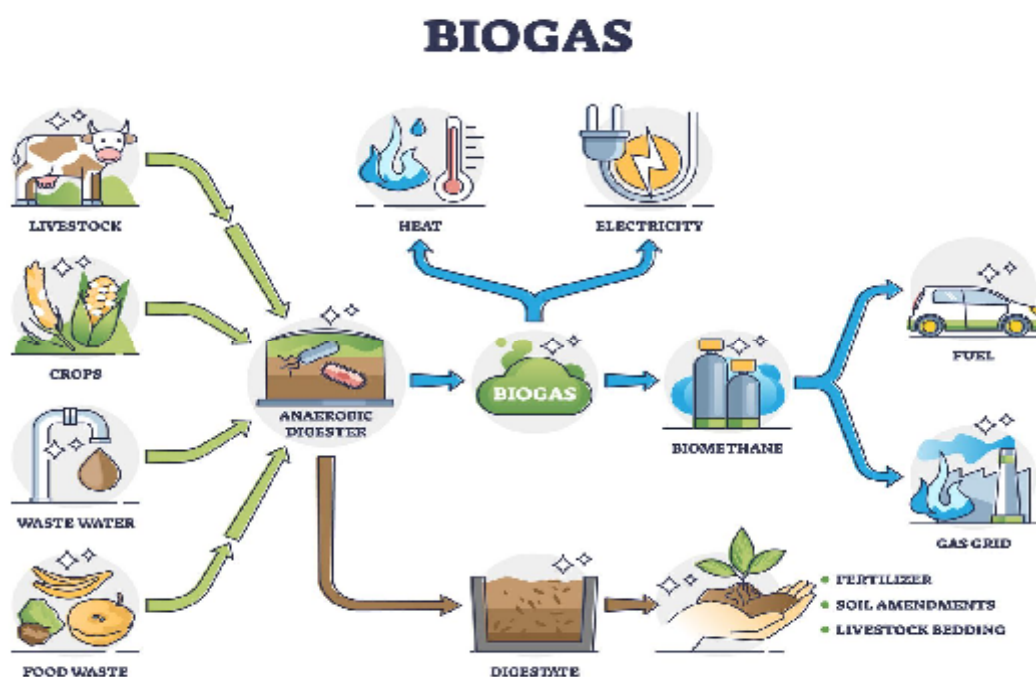
Biogas a renewable fuel that is produced when organic matter, such as food or animal waste, is broken down by microorganisms in the absence of oxygen. This process is called anaerobic digestion. For this to take place, the waste material needs to be enclosed in an environment where there is no oxygen

10.1 INTRODUCTION

Anaerobic digestion is a process that breaks down organic matter into simpler chemical components in the absence of oxygen. This process has proved to be very effective to treat organic wastes for minimizing environmental pollution. The common organic wastes are listed as follows:

1. Sewage sludge
2. Organic farm wastes
3. Municipal solid wastes
4. Organic industrial and commercial wastes
5. Forests and agricultural wastes

The digestion process itself takes place in digester, which is classified in terms of temperature, water content of feedstock and the number of stages (single or multi-stage). The by-products of anaerobic digestion, namely biogas and digestate, can be used to create a source of income.



10.2 BIOGAS AND ITS COMPOSITION

The main component of biogas is methane (CH_4) which is popularly known as biogas, gohar gas, clear gas, etc. it is clean non-polluting and low-cost gas.

<i>Substances</i>	<i>Symbol</i>	<i>Percentage</i>
Methane	CH ₄	50 – 70
Carbon Dioxide	CO ₂	30 – 40
Hydrogen	H ₂	5 – 10
Nitrogen	N ₂	1 – 2
Water vapour	H ₂ O	0.3
Oxygen	O ₂	0 – 2
Hydrogen Sulphide	H ₂ S	Traces

Review questions:

1. List application of biomass gasifier.
2. Define biogas
3. List the composition of biogas

Session 28

10.3 ANAEROBIC DIGESTION

It is a biological process that produces a gas (commonly known as biogas) in the absence of oxygen and has major components of methane (CH₄) and carbon dioxide (CO₂).

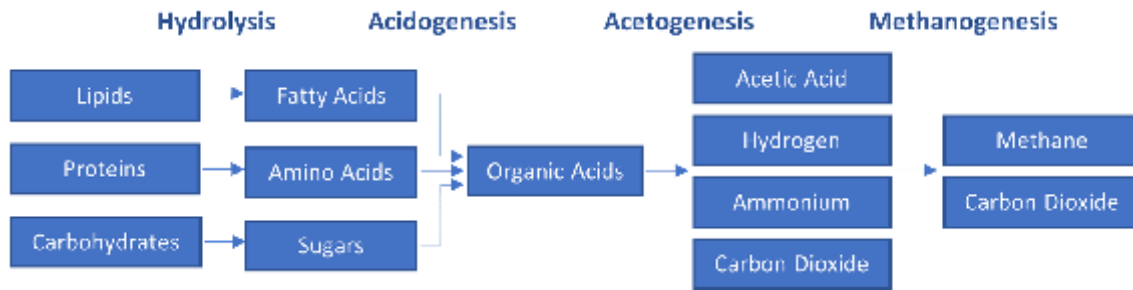
Anaerobic digestion of methane gas production is a series of processes in which microorganism break down biodegradable material in the absence of oxygen which completes through following steps:

1. In the first step, the **organic matter** (e.g. plants residues, human and animal wastes and residues) is decomposed (hydrolysis) to break down the organic material into **usable-sized molecules such as sugar**.
2. Conversion of decomposed matter into **organic acids** is the second step.
3. Finally, organic acids are converted to **biogas (methane gas)**.

10.3.1 Process Stages of Anaerobic Digestion

The biological and chemical stages of anaerobic digestion are shown in Figure 10.1. These are divided into the following four main stages:

1. Hydrolysis
2. Acedogenesis
3. Acetogenesis
4. Methanogenesis



The four main stages are explained as follows.

10.3.1.1 Hydrolysis

The process of breaking large biomass organic chains into their smaller constituent parts such as sugar, fatty acids, and amino acids and dissolving the smaller molecules into solution is called hydrolysis. This process assists bacteria in anaerobic digesters to access the energy potential of the material.

10.3.1.2 Acidogenesis

Acidogenesis is the biological process in which the remaining components are broken down by acidogenetic (fermentative) bacteria. It creates volatile fatty acids together with ammonia, carbon dioxide, and hydrogen sulphide, and other by-products.

10.3.1.3 Acetogenesis

In this stage of anaerobic digestion, simple molecules created through the acidogenesis phase are further digested to produce more acetic acid, carbon dioxide, and hydrogen.

10.3.1.4 Methanogenesis

Finally, the process of biogas production is completed by methanogenesis. In this stage of anaerobic digestion, the methanogens use intermediate products of the preceding stages and convert them into methane, carbon dioxide, and water which makes most of the biogas emitted from the system. Methanogenesis is sensitive to both high and low pH values.

A simplified generic chemical equation for the overall processes outlined earlier is as follows:

$$C_6H_{12}O_6 \rightarrow 3CO_2 + 3CH_4$$

The remaining indigestible material cannot be used by microbes and any dead bacterial remains constitute the digestate.

10.4 BIOGAS PRODUCTION

Anaerobic processes either occur naturally or created in a controlled environment, namely a biogas plant in which organic wastes are put in an airtight container called digester to perform anaerobic digestion process.

10.4.1 Construction Parts of Biogas Plants

Figure 10.2 shows various parts of typical biogas plant. It is a brick and cement structure having the following five sections:

1. Mixing tank
2. Digester tank
3. Dome or gas holder
4. Inlet chamber
5. Outlet chamber

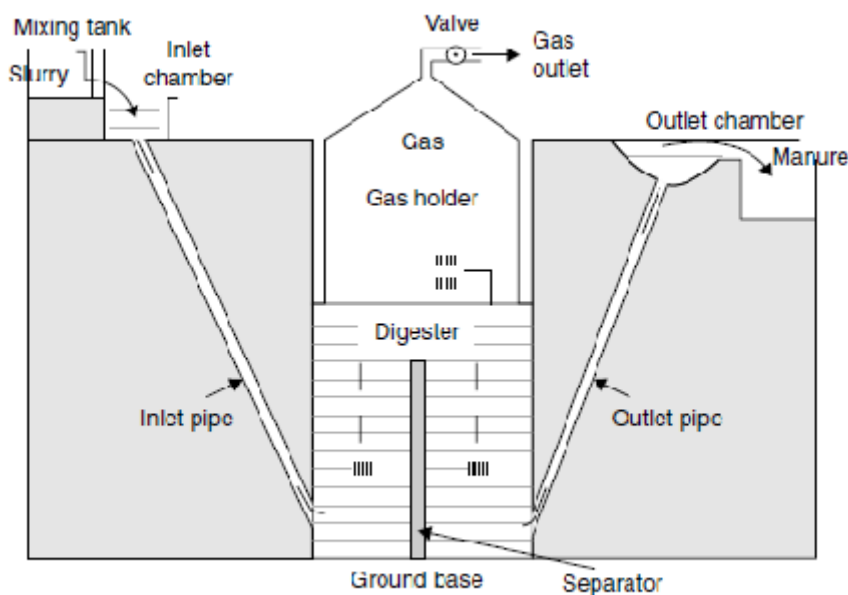


Figure 10.2 A typical biogas plants

10.4.1.1 Mixing Tank

It is the first part of biogas plants located above the ground level in which the water and cow dung are mixed in equal proportions (the ratio of 1:1) to form the slurry that is fed into the inlet chamber.

10.4.1.2 Digester Tank

It is a deep underground well-like structure and is divided into two chambers by a partition wall in between. It is the most important part of the cow dung biogas plants where all the important chemical processes or fermentation of cow dung and production of biogas takes place. The digester is also called as fermentation tank. It is cylindrical in shape and made up of bricks, sand, and cement built underground over the solid foundation. Two openings are provided on the opposite sides and at the specified height of digester for inflow of fresh cow dung slurry and outflow of used slurry as manure.

The two long cement pipes are used as follows:

1. Inlet pipe opening into the inlet chamber for inputting the slurry in digester tank.

2. Outlet pipe opening into the overflow tank (outlet chamber) for the removal of spent slurry from the digester tank. A separator is also placed in the middle of digester tank to improve effective fermentations of feedstock.

10.4.1.3 Dome or Gas Holder

The hemispherical top portion of the digester is called dome. It has fixed height in which all the gas generated within the digester is collected. The gas collected in the dome exerts pressure on the slurry in the digester. The dome or gas holder is made either fixed dome or floating dome type. Cement and bricks are used in the construction of fixed dome, and it is constructed using approximately at the ground surface.

Floating dome type is an inverted steel drum resting on the digester above the ground surface. The drum floats over the digester and moves up and down with biogas pressure.

10.4.1.4 Inlet Chamber

The cow dung slurry is supplied to the digester of the biogas plant via inlet chamber, which is made at the ground level so that the slurry can be poured easily. It has bell mouth sort of shape and is made up of bricks, cement, and sand.

10.4.1.5 Outlet Chamber

The digested slurry from the biogas plants is removed through the outlet chamber. The opening of the outlet chamber is also at the ground level. The slurry from the outlet chamber flows to the pit made especially for this purpose.

10.4.1.6 Gas Outlet Pipe and Valve

The gas holder has an outlet at the top which could be connected to gas stoves for cooking or gas-lighting equipment or any other purpose. Flow of the gas from the dome via gas pipe can be controlled by valve. The gas taken from the pipe can be transferred to the point of use.

10.4.1.7 Foundation

The foundation forms the base of the digester where the most important processes of biogas plant occur. It is made up of cement, concrete, and bricks strong enough so that it should be able to provide stable foundation for the digester walls and be able to sustain the full load of slurry filled in it. The foundation should be waterproof so that there is no percolation and leakage of water.

10.4.2 Working of Biogas Plant

1. Cattle dung and water mixed in equal proportion and poured into inlet chamber.
2. Through inlet pipe slurry enters the fermentation tank, digestion takes place biogas is formed. Around 60 days anaerobic bacteria will ferment the slurry.
3. This gas is accumulated in gas holder. Outlet pipe will be closed due to pressure in holder it pushes slurry to outlet chamber.
4. The biogas creates a bubble in inlet and outlet chamber during escape by understanding this gas valve to be opened for various application.
5. As the slurry is removed fresh slurry to be added. Size of plant decides the capacity of plant.

10.4.3 Types of Biogas Plants

Fixed dome and floating dome construction are the two types of biogas plants. Based on these types, several biogas plant models are developed.

10.4.3.1 Fixed Dome Type

Schematic of a fixed dome biogas plant is given in Figure 10.3. It consists of following parts

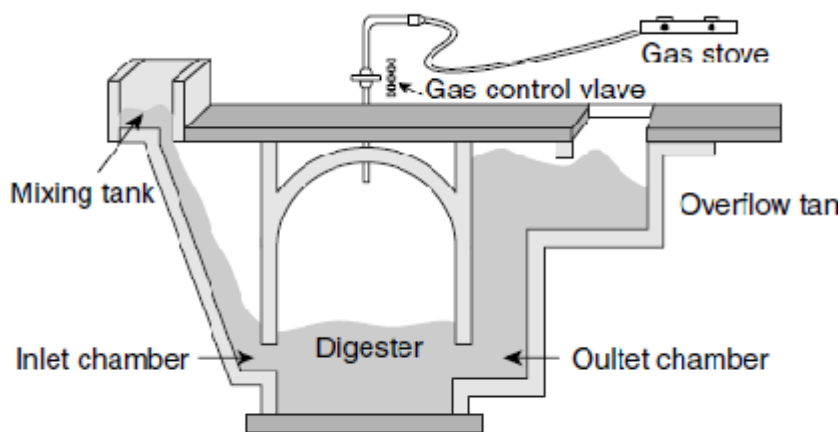


Figure 10.3 Fixed dome type biogas plant

1. Mixing tank: In mixing tank, the water and cattle dung are mixed together thoroughly in the ratio of 1:1 to form the slurry.
2. Inlet chamber: The mixing tank opens underground into a sloping inlet chamber.
3. Digester: Digester is a huge tank with a dome type ceiling. The ceiling of the digester has an outlet with a valve for the supply of biogas. The inlet chamber opens from below into the digester tank. The digester opens from below into an outlet chamber which is opened from the top into a small overflow tank.

10.4.3.1.1 Working Principle

The various forms of organic biodegradable biomass are **collected and mixed with equal amount of water properly in the mixing tank to form slurry**. The slurry is fed into the digester tank through inlet chamber and pipe, and the digester is partially filled by about half of its height. The feeding of slurry is then discontinued for about 60 days when anaerobic bacteria present in the slurry decomposes or ferments the biomass in the presence of water. Biogas is then formed and starts accumulating in the upper dome area of the biogas plants, and the pressure is exerted on the spent slurry to force it flow into the outlet chamber. Finally, the spent slurry overflows into the overflow tank from where it is manually removed and used as manure for agricultural crops and plants. Gas control valve at the top of dome is opened

partially or fully to supply required gas for applications. A functioning plant is fed continuously with the prepared slurry to obtain a continuous supply of biogas

10.4.3.1.2 Advantages of fixed dome-type biogas plant are as follows:

1. The costs of a fixed dome biogas plant are relatively low as compared to floating dome type.
2. It is simple in construction as no movable dome exists.
3. It is made up of concrete, bricks, and cements and long life of the plant (20 years or more) can be expected.
4. Underground and almost ground surface dome construction saves space and protect from physical damage to the plant.
5. The anaerobic digestion processes in the digester are little influenced by temperature fluctuation in day and night.

10.4.3.1.3 Disadvantages of biogas plant are as follows:

1. Porosity and cracks in plant walls is the major drawbacks.
2. Maintenance is rather difficult.

10.4.3.2 Floating Type

The floating gas holder type of biogas plant is shown in Figure 10.4. The construction and working principle of this biogas plants is similar to fixed dome type **except that gas holder tank is made up of steel and placed on the top of digester circular tank and is movable up and down also shown in Figure**

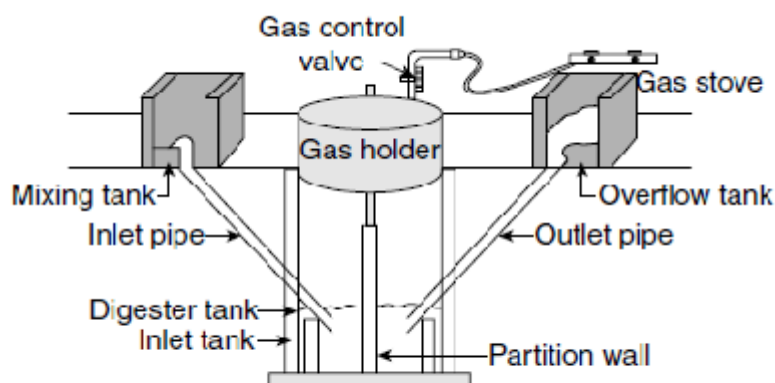


Figure 10.4 Floating dome-type biogas plant

10.4.3.2.1 Advantages Floating dome-type biogas plant has the following advantages:

1. Very efficient
2. Simple maintenance scheduling possible

10.4.3.2.2 Disadvantages Floating dome-type biogas plant has the following disadvantages:

1. Expensive
2. Steel drum may rust
3. Requires regular maintenance

BENEFITS OF BIOGAS

1. **Production of energy**- the calorific value of biogas is 6KWh/m³. small and medium units are used for the purpose of cooking and lighting purpose. Large unit is used for power generation.
2. **Transformation of organic waste into high quality organic fertilizer**- also called as fertilizer producer, the fertilizer comes out has 3 times more nitrogen compared to open air digestion. Since it is closed type nitrogen is preserved in it no chance of escaping.
3. **Health benefits of biogas plant and improvement of hygienic condition**- respiratory diseases, illness, eye infection, asthma are avoided where the plant is been established. hygienic cooking is possible and many harmful organic organisms are killed inside the plant.
4. **Reduction of workload**: work load for women is reduced in collecting wood from distance places, cleaning and firing it. Home remains free from smoke and dust.
5. **Environmental advantages**: protection of soil, air, forest, and water i.e helps in deforestation.
6. **Global environmental benefits of biogas technology**- capturing CH₄ will reduce global warming. CO₂emmission is also reduced

FACTORS AFFECTING THE SELECTION OF A MODEL OF A BIOGAS PLANT

1. **Cost**: the initials and maintenance cost to be low as possible
2. **Simplicity in design**: The design should be simple for ease of operation. As rate of skilled person using the plant is less.
3. **Durability**: longer lifespan of plant is required where in people will adopt the technology and use.
4. **Suitability for use with available raw materials**: based on raw material used it should be designed.
5. **Input and output use frequency**: input and output design must be in particular size and potential as it frequently used.

Review questions:

1. Define anaerobic digest
2. Benefits of biogas
3. Types of biogas plant

Session 29

BIOGAS PLANT FEEDS AND THEIR CHARACTERISTICS

Any type of biodegradable material can be used as a source but for economical reason few are preferred. **Cattle dung is mostly used as input.** Based on input gas production will vary. Carbon nitrogen ratio (C/N) ratio-ratio of carbon and nitrogen present in a organic matter. If **ratio is ranging from 20-30 then it is considered as optimum for anaerobic digestion.** A pH value should not exceed 8.5 if it increases it shows toxic effect. C/N ratio of cattle dung is 24 and human is 8. These may be mixed to bring it into optimum condition.

Table 10.2 Gas Production Potential of Various Types of Dung

Types of Dung	Gas Production Per Kg Dung (m ³)
Cattle (cows and buffaloes)	0.023–0.040
Pig	0.040–0.059
Poultry (Chickens)	0.065–0.116
Human	0.020–0.028

Table 10.3 C/N Ratio of a Few Commonly Used Materials

Raw Materials	C/N ratio
Duck dung	8
Human excreta	8
Chicken dung	10
Goat dung	12
Pig dung	18
Sheep dung	19
Cow dung/buffalo dung	24
Water hyacinth	25
Elephant dung	43
Straw (maize)	60
Straw (rice)	70
Straw (wheat)	90
Saw dust	above 200

Advantage:

1. clean fuel
2. No residue, smoke, dust are produced
3. · Non polluting
4. Health and Economical benefits.
5. Provide nutrient rich manure

Limitations:

1. high initial cost
2. availability of raw material continuously
3. social acceptability.
4. Maintenance and repair cost is high

Uses:

1. Used as domestic fuel
2. Fuel for motive power
3. Used for electricity generation

Tidal Energy

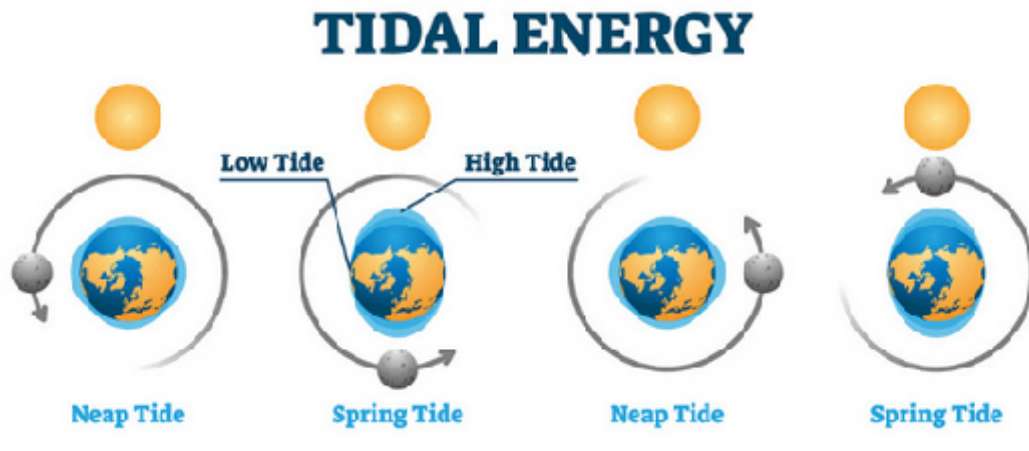
Tides are periodic rises and falls of large bodies of water. Gravity is one major force that creates tides. In 1687, Sir Isaac Newton explained that ocean tides result from the gravitational attraction of the sun and moon on the oceans of the earth. Spring tides are especially strong tides that occur when the earth, the sun, and the moon are in a line. The gravitational forces of the moon and the sun both contribute to the tides. Spring tides occur during the full moon and the new moon. Neap tides are especially weak tides. They occur when the gravitational forces of the moon and the sun are perpendicular to one another with respect to the earth. Neap tides occur during quarter moons.

TIDAL ENERGY RESOURCE

Tides are the waves caused **due to the gravitational pull of the moon and the sun** (although its pull is very low). The rise of seawater is called high tide and fall in seawater is called low tide and this process of rising and receding of water waves happen twice a day and cause enormous movement of water. Thus, enormous rising and falling movement of water is called tidal energy, which is a large source of energy and can be harnessed in many coastal areas of the world. Tidal dams are built near shores for this purpose in which water flows during high tide and water flows out of dam during low tides. Thus, the head created results in turning the turbine coupled to electrical generator.

Tidal energy has been developed on a commercial scale among the various forms of energy contained in the oceans. **When the moon, the earth, and the sun are positioned close to a**

straight line, the highest tides called spring tides occur. When the earth, moon, and sun are at right angles to each other (moon quadrature), the lowest tides called **neap tides occur.** The water mass moved by the moon's gravitational pull when moon is very close to ocean and results in dramatic rises of the water level (tide cycle). The tide starts receding as the moon continues its travel further over the land, away from the ocean, reducing its gravitational influence on the ocean waters (ebb cycle).



TIDAL ENERGY AVAILABILITY

Gravitational forces between the moon, the sun, and the earth cause the rhythmic rise and fall of ocean waters throughout the world. Those result in tide waves. The moon exerts more than twice as great a force on the tides as the sun due to its much closer position to the earth. As a result, the tide closely follows the moon during its rotation around the earth, creating diurnal tide and ebb cycles at any ocean surface. The amplitude or height of the tide wave is very small in the open ocean where it measures several centimetres in the centre of the wave distributed over hundreds of kilometres. However, the tide can increase dramatically when it reaches continental shelves, bringing huge masses of water into narrow bays, and river estuaries along a coastline. For instance, the tides in the Bay of Fundy in Canada are the greatest in the world, with amplitude between 16 and 17 m near shore. Table 11.1 gives ranges of amplitude for some locations with large tides.

Table 11.1 Highest Tides (Tide Ranges) of the Global Ocean

Country	Site	Highest Tide Range (m)
Canada	Bay of Fundy	16.2
England	Severn Estuary	14.5
France	Port of Granville	14.7
France	La Rance	13.5
Argentina	Puerto Rio Gallegos	13.3
Russia	Bay of Mezen (White Sea)	10.0
Russia	Penzhinskaya guba	13.4
India	The Gulf of Cambay, Gujarat	11
India	Gulf of Kutch, Gujarat	8
India	The Ganges Delta in the Sundarban, West Bengal	5

Source: NOAA Federal

Table 11.2 provides glimpses of few potential sites for tidal power generation.

Table 11.2 A Few Potential Sites for Tidal Power Generation

Country	Site	Average Tide Height (m)	Basin Area (m ²)	Estimated Power Potential (MW)
Argentina	San-Jose	6.0	780	7,000
Australia	Secure	8.4	130	570
Australia	Walcoti	8.4	260	1,750
Korea	Carolina Bay	4.7	90	480
Russia	Mezen	5.66	2,640	15,000
Russia	Tugur	5.38	1,080	6,790
UK	Severn	8.3	490	6,000
UK	Mercy	8.4	60	700
USA	Cook Inlet	4.35	3,100	18,000
USA	Passamaquoddy	5.55	300	400

Source: NOAA Federal

Review questions:

1. Advantages and disadvantages of biogas
2. Define tidal energy.
3. Tidal energy resources.

Session 30

TIDAL POWER GENERATION IN INDIA

Important site location and estimated power potential of a few Indian tidal energy plant is given in Table 11.3.

Table 11.3 Indian Tidal Energy Plant

Site Location	Tide Heights (m)	Estimated Power Potential (MW)
The Gulf of Cambay, Gujarat	11 (6.7 av)	7,000
Gulf of Kutch, Gujarat	8 (5.23 av)	12,000
The Ganges Delta in the Sundarban, West Bengal	5 (2.97)	8,000

Nevertheless, the possibility of developing tidal power scheme in India may be examined in the following all aspects:

1. Economic aspects of tidal power schemes when compared to the conventional schemes.
2. Problems associated with the construction and operation of plant.
3. Problems related to the hydraulic balance of the system to minimize the fluctuation in the power output.
4. Environmental effects of the schemes

11.5 LEADING COUNTRY IN TIDAL POWER PLANT INSTALLATION

Worldwide installed capacity of few countries are approximately shown in Table 11.4.

Table 11.4 Installed Tidal Power Capacities of Few Countries

Country	Site Location	Installed Capacity (MW)
France	La Rance	24 bulb-type turbines, each of 10 MW rating. Total 240 MW
UK	The Severn Barrage	A total of 214 turbines each of 40 MW rating. Total = 8,560 MW
Russia	Kislaya Guba	0.4 MW
Canada	Annapolis	18 MW
China		3.9 MW

ENERGY AVAILABILITY IN TIDES

Potential energy and kinetic energy are the two energy components of energy of the tide waves. The potential energy is the work done in lifting the mass of water above the ocean surface.

This energy can be calculated as

$$E = \rho g A \int z dz = 0.5 \rho g A h^2 \quad (11.1)$$

where E = the energy; g = acceleration of gravity; ρ = the seawater density (which equals its mass per unit volume); A = sea area under consideration; z = vertical coordinate of the ocean surface; and h = the tide amplitude.

Taking an average of the product of $(\rho g) = 10.15 \text{ kN/m}^3$ for seawater, energy for a tidal cycle per m^2 of ocean surface can be approximated as:

$$E = 1.4 h^2 \quad (\text{Watt-hour}) \quad (11.2)$$

or

$$= 5.04 h^2 \quad (\text{kJ}) \quad (11.3)$$

Since extracting all the available stream power could be environmentally damaging, it is necessary to use a factor that expresses the usable power percentage with apparently no damaging consequences. It is called α (significant extraction factor) that may vary from 0.2 to 0.6.

The kinetic energy (KE) of the water mass (m) is its capacity to do work by virtue of its velocity (V). It is defined by

$$\text{KE} = 0.5mV^2 \quad (11.4)$$

The total energy of tide waves equals the sum of its potential and kinetic energy components.

Estimation and understanding of the potential energy availability of the tides are key for designing conventional tidal power plants using water dams for creating artificial upstream water heads. Such power plants exploit the potential energy of vertical rise and fall of the water.

The kinetic energy of the tide has to be known for designing other types of tidal power plants (like Soating), which harness energy from tidal currents or horizontal water. They do not involve the installation of water dams.

11.6.1 Calculation of Tidal Power

Potential tidal power can be reckoned based on a mathematical calculation. Let us assume that the surface area of the reservoir as stable between the full stored water level and the emptied floor, the energy produced by the ebbing water could be expressed as

$$d(w) = \rho g h d(v) = \rho g A h d(h) \quad (11.5)$$

Here, $d(w)$ = energy unit; ρ = density of seawater (about $[1.02\text{--}1.04] \times 10^3 \text{ kg/m}^3$); g = acceleration of gravity (9.8 m/s^2); A = surface area of the reservoir (m^2) assumed as a constant from high tide to low tide; h = instant water level height (m); v = volume of reservoirs (m^3).

Therefore, its power could be written as

$$P = \int d(w) / \int dt = \rho g A H^2 / 2T \quad (11.6)$$

Here, P = potential power (W); T = tidal period (s); and H = tidal ranges (m).

Let us assume $\rho = 1.04 \times 10^3 \text{ kg/m}^3$ and $T = 6 \text{ h}$.

This formula can be simplified as $P = 0.226AH^2 \text{ (W)}$ (11.7)

Example 11.1

For a typical tidal power plant shown in Figure P11.1, the basin area is $25 \times 10^6 \text{ m}^2$. The tide has a range of 10 m. However, turbine stops working when the head on it falls below 2 m. Assume that density of seawater is $1,025 \text{ kg/m}^3$, acceleration due to gravity is 9.81 m/s^2 , combined efficiency of turbine and generator is 75%, and period of energy generation is 6 h and 12.5 min. Calculate:

1. Work done in filling or emptying the basin.
2. Average power
3. The energy generated in one filling process.

Solution

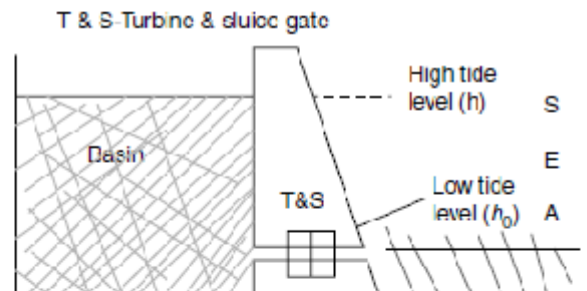


Figure P11.1 Single-basin tidal plant

1. Total work done in filling or emptying the basin

h_0

$$W = \int \rho g A h d(h) = \frac{1}{2} \rho g A (h^2 - h_0^2)$$

h

$$= \frac{1}{2} \times [1,025 \times 9.81 \times 25 \times 10^6 \times (10^2 - 2^2)] \\ = 17.6 \times 10^{12}$$

2. Average power

$$P_{av} = W/t = 17.6 \times 10^{12} / 22,350 = 787.32 \times 10^6 \text{ W}$$

3. Energy generated

$$E = 0.75 \times 787.32 \times 10^6 \times 3,600 / 100 = 2.123 \times 10^9 \text{ kWh}$$

Tidal Stream Generator

It is often referred to as a tidal energy converter (TEC). It is a machine that extracts kinetic energy from moving masses of water in particular tides. A TEC device extracts energy from a tidal flow much in the same way that a windmill extracts energy from the wind. The following equation can be used to calculate the power output of either devices,

$$P = \frac{1}{2} \rho A \eta_T V^3 \quad (11.8)$$

where P – power (W); ρ – density of seawater (kg/m^3); A – capture area (m^2); η_T – combined efficiency from water to electric wire; and V – flow speed (m/s).

The abovementioned power calculation is valid, provided the capture area of the tidal energy conversion device(s) is small in comparison to the cross-sectional area of the channel. The significant difference between wind and tidal power calculations is the typical range of flow speeds and the density of the fluid.

Review questions:

1. Formula for tidal energy
2. Formula for tidal power
3. Tidal energy availability

Session 31

TIDAL POWER BASIN

The basin system is the most practical method of harnessing tidal energy. It is created by enclosing a portion of sea behind erected dams. The dam includes a sluice that is opened to allow the tide to flow into the basin during tide rise periods and the sluice is then closed. When the sea level drops, traditional hydropower technologies (water is allowed to run through hydro turbines) are used to generate electricity from the elevated water in the basin.

Single-basin System

Single water reservoir is closed off by constructing dam or barrage. Sluice (gate), large enough to admit the water during tide so that the loss of head is small, is provided in the dam. The single-basin system has two configurations, namely:

1. **One-way single-basin system:** The basin is filled by seawater passing through the sluice gate during the high tide period. When the water level in the basin is higher than the sea level at low tide period, then power is generated by emptying the basin water through turbine generators. This type of systems can allow power generation **only for about 5 h** and is followed by the refilling of the basin. Power is generated till the level of falling tides coincides with the level of the next rising tide.

2. **Two-way single basin:** This system allows power generation from the water moving from the sea to the basin, and then, at low tide, moving back to the sea. This process requires bigger and more expensive turbine.

Single-basin system has the drawbacks of intermittent power supply and harnessing of only about 50% of available tidal energy

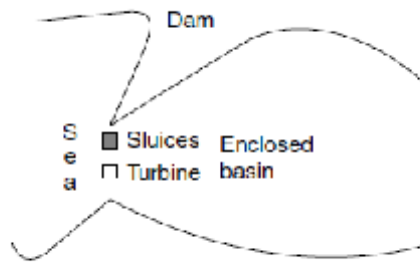


Figure 11.1 Single-basin system

Two-basin Systems

The two basins close to each other, operate alternatively. One basin generates power when the tide is rising (basin getting filled up) and the other basin generates power while the tide is falling (basin getting emptied). The two basins may have a common power house or may have separate power house for each basin. In both the cases, the power can be generated continuously. The system could be thought of as a combination of two single-basin systems, in which one is generating power during tiding cycle, and the other is generating power during emptying.

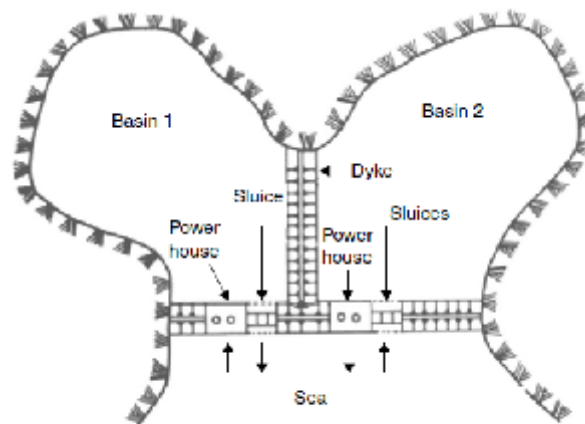


Figure 11.2 Two-basin system

Co-operating Two-basin Systems

This scheme consists of two basins at different elevation connected through the turbine. The sluices in the high- and low-level basin communicate with seawater directly, as shown in Figure 11.3. The high-level basin sluices are called the inlet sluices and the low level as outlet sluices. The basic operation of the scheme is as follows:

1. The rising tide fills the high-level basin through the sluiceways.
2. When the falling seawater level is equal to the water level in the high-level basin, the sluiceways are closed to prevent the outflowing high-level basin water back to the sea.
3. The water from high-level basin is then allowed to flow through the turbine generators to the low-level basin.
4. When the falling seawater level becomes lower than the rising water level in the low-level basin, the sluiceways are opened to allow water to flow into the sea from the low-level basin. This process continues until the water level in the low-level basin equals to the rising sea level. Then, the sluiceways are closed to prevent the filling of low-level basin from the seawater.

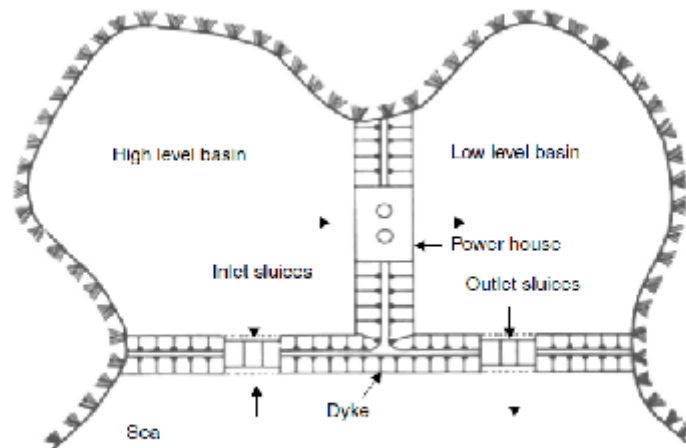


Figure 11.3 Co-operating two-basin systems

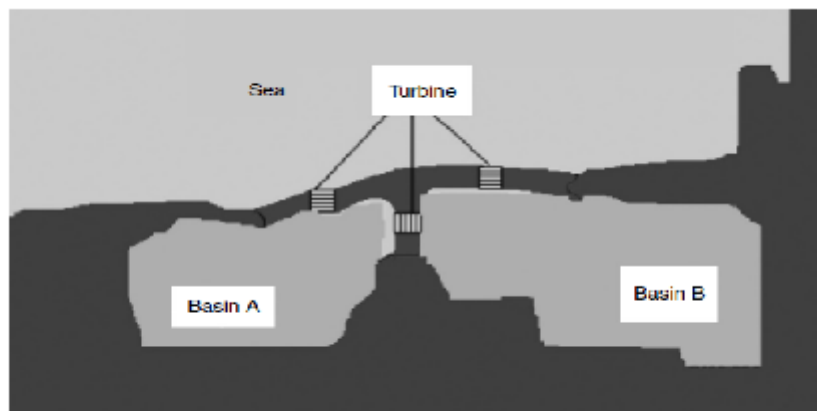


Figure 11.4 Co-ordinating two basin systems

Figure 11.4 gives another schematic diagram of co-coordinating two-basin tidal power stations. With two basins, one is filled at high tide and the other is emptied at low tide. Turbines are placed between the basins and between the basin and the sea. These two basin systems allow

continuous power generation. However, they are very expensive to construct due to the cost of the extra length.

TURBINES FOR TIDAL POWER

The bulb-type turbine shown in Figure 11.5 consists of a steel shell completely enclosing the generator that is coupled to the turbine runner.

The turbine is mounted in a tube within the structure of the barrage, and the whole machine being always submerged. When the power demand on the system is low during the rising tides, the unit operates as a pump to transfer water from sea to the basin. When the load on this system is high, the unit will work as a generator, and deliver the stored energy that is a valuable additional input to the system.

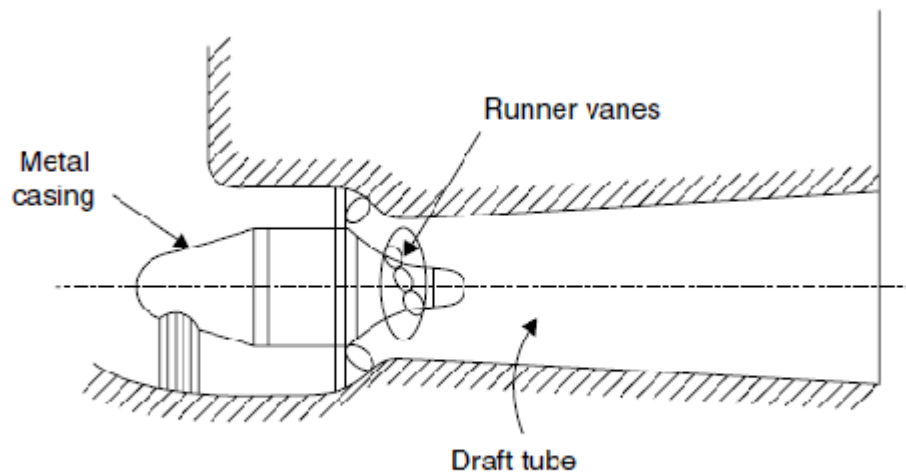


Figure 11.5 Bulb-type turbine

11.8.2 Commercial Status of Tidal Stream Devices (as on 2009)

Table 11.5 Status of Tidal Stream Devices

Company	Class	Technology	Country	Year	Stage
Aqua Marine Power	Tidal	Horizontal Axis Turbine	UK	2007	Prototype
Vendant Power	Tidal	Horizontal Axis Turbine	US	2000	Commercial
Marine Current Turbines	Tidal	Horizontal Axis Turbine	UK	2000	Commercial
SMD Hydrovision	Tidal	Horizontal Axis Turbine	UK	2003	Prototype
Open-Hydro	Tidal	Open Centre Turbine	Ireland	2006	Pre-commercial
Hammerfest Strom	Tidal	Horizontal Axis Turbine	Norway	2007	Pilot

Review questions:

1. Types of tidal powder basin
2. How two basin s work
3. How bulb type turbine work.

Session 32

ADVANTAGES AND DISADVANTAGES OF TIDAL POWER

The following are the advantages of tidal power:

1. About two-third of earth's surface is covered by water, there is scope to generate tidal energy on large scale
2. Techniques to predict the rise and fall of tides as they follow cyclic fashion and prediction of energy availability is well established.
3. The energy density of tidal energy is relatively higher than other renewable energy sources.
4. Tidal energy is a clean source of energy and does not require much land or other resources as in harnessing energy from other sources.
5. It is an inexhaustible source of energy.
6. It is an environment friendly energy and does not produce greenhouse effects.
7. Efficiency of tidal power generation is far greater when compared to coal, solar, or wind energy. Its efficiency is around 80%.
8. Even though capital investment of construction of tidal power is high, running and maintenance costs are relatively low.
9. The life of tidal energy power plant is very long.

The following are the disadvantages of tidal power:

1. Capital investment for construction of tidal power plant is high.
2. Only a very few ideal locations for construction of plant are available and they too are localized to coastal regions.
3. Unpredictable intensity of sea waves can cause damage to power generating units.
4. Aquatic life is influenced adversely and can disrupt the migration of fish.
5. The energy generated is not much as high and low tides occur only twice a day and continuous energy production is not possible.
6. The actual generation is for a short period of time. The tides only happen twice a day so electricity can be produced only for that time, approximately for 12 h and 25 min.
7. This technology is still not cost effective and more technological advanc

to make it commercially viable

Review questions:

1. Advantages of tidal energy
2. Disadvantages of tidal energy.